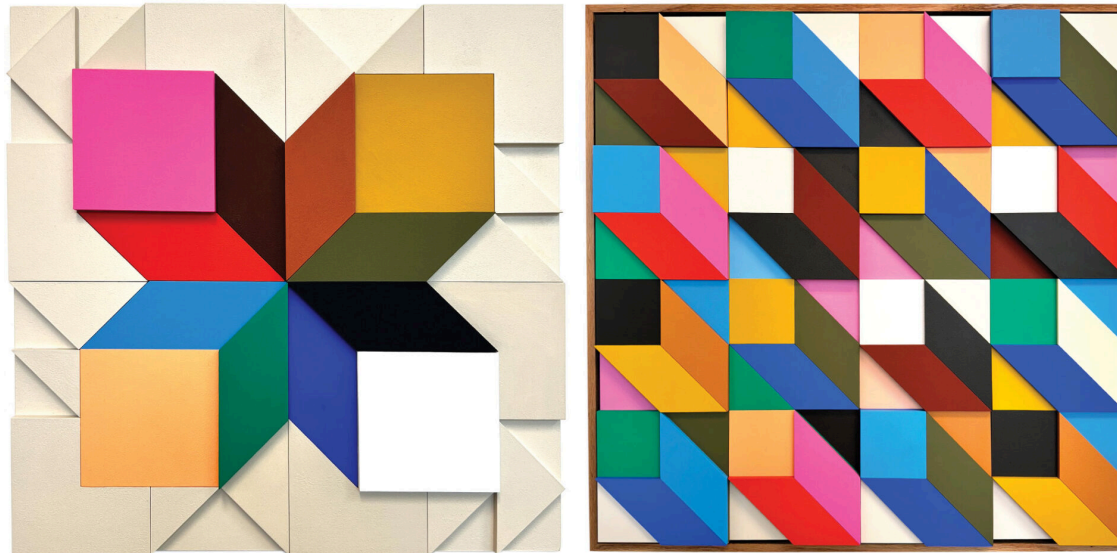


Innovation

Why You Need Systems Thinking Now

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Summary. Traditional innovation approaches—breakthrough and design thinking—often ignore the complex ripple effects they produce in interconnected systems. In a world facing multifaceted challenges like climate change and sustainability, a systems-thinking approach offers a more resilient and holistic path forward. Systems thinking emphasizes understanding interdependencies, redefining



problems iteratively, and engaging diverse stakeholders to co-create solutions. This article outlines a four-step framework: (1) Define a desired future state. (2) Reframe problems so that they resonate across stakeholders. (3) Focus on flows and relationships over discrete products. And (4) implement small nudges to gradually shift the system. Examples from Maple Leaf Foods, Co-operators Insurance, and the CSA Group illustrate how this approach can realign business models for long-term sustainability. While not a replacement for other innovation methods, systems thinking addresses their limitations in tackling “wicked problems.” It encourages companies to anticipate unintended consequences, build coalitions, and adaptively guide transformation in complex environments. [close](#)

Business has made huge strides in advancing economic and social prosperity in recent decades through innovative technologies and new ways of working. But many of those innovations have costs. For example, plastics are used to make many convenient and low-cost consumer products, but they create huge problems for both terrestrial and marine ecosystems and find their way into thousands of kinds of animals, including humans. Fracking technology has kept the price of oil low, but it has harmful effects on water resources and contributes to air pollution and other environmental problems. In finance, credit default swaps were invented to help investors hedge credit risk, but they ended up precipitating the 2008 global financial crisis.



Although the unforeseen consequences of innovation can to some extent be mitigated retrospectively through regulation and tax policy, we believe that a better approach is to avoid them in the first place by thinking more carefully about how innovation is done.

In this article we'll look at the strengths and weaknesses of the two dominant approaches that businesses apply to innovation—*breakthrough thinking* and *design thinking*—which often produce socially and environmentally dysfunctional outcomes in complex systems. To avoid them, innovators should apply *systems thinking*, a methodology that has been around for decades but is rarely used today. It addresses the fact that in the modern economy every organization is part of a network of people, products, finances, and data, and changes in one area of the network can have side effects in others. For example, recent attempts by the U.S. government to impose tariffs on foreign imports have had ripple effects on the supply chains of major products like cars and iPhones, whose components are sourced



from multiple countries. The tariff plans have also led to a spiral of complex and unpredictable reactions in financial markets.

Systems thinking helps predict and solve problems in dynamic, interconnected environments. It's especially relevant to innovation for sustainability challenges. Electric vehicles, for example, have attracted a lot of investment, notably in China, because they are seen as a green technology. But their net effect on carbon emissions is highly contingent on how green a country's power supply is. What's more, their technology requires raw materials whose processing is highly polluting. Solar panels also look like an environmental silver bullet, but the rapidly growing scale of their manufacturing threatens to produce a tsunami of electronic waste. Truly sustainable technology solutions for environmental challenges require a systems-led approach that explicitly recognizes that the benefits of an innovation in one part of the planet's ecology may be outweighed by the harm done elsewhere.

In this article we'll offer a set of principles that can guide companies in applying systems thinking to complex problems. Our advice is based on our work at Innovation North, a research-practice initiative at Ivey Business School in London, Canada, in which we collaborated with over 30 major companies on more than a dozen large projects and numerous agile sprints. Drawing



on insights from that work, we'll compare the pros and cons of the breakthrough, design, and systems thinking approaches.

Three Modes of Innovation

Perhaps the most popular approach to business innovation is breakthrough thinking. It's the 10x, winner-takes-all model characterized by Mark Zuckerberg's mantra to "move fast and break things." Here the innovator, typically an entrepreneur armed with a new piece of technology, ignores existing relationships, norms, and even laws to create a better product.

For example, when Google set out to improve its search engine, it didn't ask permission to make use of people's personal information. Uber didn't work with local authorities and taxi companies when it launched its services in new cities—it just did it. The mindset of these and other Silicon Valley companies is to move forward and worry about the consequences later. Like Alexander the Great, they make progress by slicing through the Gordian knot of a complex problem, rather than attempting to unpick it.

When it works, breakthrough thinking delivers huge rewards for its practitioners. But it can also create a lot of collateral damage. When Google rode roughshod over privacy and IP laws, it became



the target of a significant public outcry and found itself facing many legal challenges. Uber put many taxi drivers out of business, and its gig workers lack many of the benefits and protections of full-time employees. Typically, the extent of damage from breakthrough thinking is highest when it is applied to “wicked problems”—those that are constantly changing and hard to define and whose solutions involve difficult trade-offs. The more complex the ecosystem around the problem, the more likely it is that the solution will impose economic and social costs. So while breakthrough thinking is ideal if the problem is clearly bounded—like getting a rocket into space—it won’t work well when it comes to, say, fixing the U.S. healthcare and education systems.





Angie Quintanilla Coates creates bright, geometric compositions that are inspired by her Mexican heritage and offer a message of optimism.

The second common innovation approach, design thinking, was popularized in the 1990s by companies like IDEO. It has become the go-to methodology for consultants and innovation teams, especially when the context is complex and features multiple



interconnected actors with divergent goals. Design thinking cuts through that complexity by focusing primarily on the users of the product or service being designed. Innovators study people in the contexts in which they will use an offering, empathize with them, divine their unarticulated needs, and then seek to reshape products and processes in ways that improve their experience. Rather than slicing through the Gordian knot, design thinkers focus on one strand in the knot (the user) and ignore everything around it.

Design thinking has a track record of success, but it too has significant downsides as the obsessive focus on the user creates knock-on problems for other parties. For example, Airbnb solved a problem for homeowners and vacationers, but it created a host of problems for local communities.

The third approach, systems thinking, emerged from the pioneering work of theoretical biologist Ludwig von Bertalanffy, computer scientist Jay Forrester, and others. It had its heyday in the 1990s with the publication of Peter Senge's bestseller *The Fifth Discipline*. Systems thinking recognizes and embraces the complexity of organizational problems, rather than seeking to simplify them. It leads to innovations that make an entire system more sustainable and resilient, avoiding the side effects and collateral damage sometimes seen in the other two approaches. At



its best, systems thinking generates more-creative solutions and greater engagement in organizational ecosystems.

Design thinkers tend to start the innovation process by focusing on the customer's problem, and breakthrough thinkers often start with the idea. But systems thinkers start by zooming out to understand the system that the innovation will be part of before they zoom in to solve the problem. That approach can lead to nonobvious answers. When facing the problem of traffic congestion, for example, a design thinker might widen or add roads. A breakthrough thinker might deploy technology to dynamically reroute drivers. A systems thinker would eschew those short-term solutions, realizing that making it easier to drive would result in a perverse incentive for people to drive more. The systems thinker might choose the counterintuitive solution of making traffic move even more slowly while increasing public transportation options. The short-term increases in congestion could be exactly what is needed to discourage people from driving and lessen congestion in the long run.

Despite its benefits, systems thinking remains the least common mode of innovation, largely because it is a slower and more complex way to solve a problem. In the traditional approach to systems thinking, the first task of the systems thinker is to identify and model all the flows, interactions, and feedback loops



in the system in question. This can be a daunting—and sometimes futile—task in a fast-changing world in which models may never accurately reflect reality. And the traditional systems-thinking innovator who spends time figuring out exactly how the Gordian knot is tied is almost guaranteed to be overtaken by a design thinker or a breakthrough thinker who simply slices through it or focuses on only one strand.

For these reasons, systems thinking may seem infeasible, even in the age of AI and quantum computing. So why are we advocating for it?

A Streamlined Approach to Systems Thinking

To reap the advantages of systems thinking you don't need to precisely model a complex adaptive system. Instead, you can develop a general understanding of critical patterns in the system and then collaborate with actors in the ecosystem to test simple ideas. The goal is to experiment, not to make wholesale, potentially catastrophic changes. A good nudge will often inspire new ideas that can transform the system.

Our streamlined approach to systems thinking has four key steps.



1. Define your desired future state. Standard practice in design thinking, and to some extent in breakthrough thinking, is to address the “job to be done” or the “unarticulated need” of the customer. Construction workers don’t want a drill; they want a hole in the wall. Music lovers don’t want to own recordings of their favorite songs; they just want to listen to them. Systems thinking, by contrast, focuses on the company’s role in some desired future state—one that cannot be achieved without changes to the many different parts of the system.

In our workshops we start by helping a company articulate its North Star—what it wants the system to deliver—and what its own role will be in that new system. This helps the company frame discussions with the stakeholders that will also have roles in the future state and whose cooperation is needed for the system—and the company—to progress toward the North Star. It also keeps innovation activities within the system from going awry. Disparate groups of individuals (within the same organization or from different organizations) can build on one another’s efforts so that innovation doesn’t become a disconnected or conflicting set of activities.

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the users of the product or service being designed.

Consider the case of Maple Leaf Foods. In 2019, then-CEO Michael McCain launched an effort to refresh the Canadian meat-processing company's purpose. He recognized that while the processed-food industry served consumers by creating cheap and tasty products, it largely ignored collateral long-term damage to human health from addictive products that contributed to obesity and increased cancer risks. Sooner or later, he believed, those societal costs would have to be addressed by the industry. What would the implications be for Maple Leaf as a meat processor?

McCain asked executives and employees to think big, think audaciously, and think about the future. They decided to become "the most sustainable protein company on Earth." No longer would Maple Leaf Foods be a meat-processing company. Its fundamental role would be to provide protein within a more environmentally sustainable and healthier food-production system.

This North Star captured Maple Leaf's new role in the food industry, how it would be fulfilled, and how the system itself would need to change. Raising livestock and processing meat



impose severe environmental costs; the increasingly urgent need to offset those costs is likely to trigger regulation and taxes, making the traditional meat-processing business far less attractive. By embracing sustainability and reframing its North Star as protein processing rather than meat processing, Maple Leaf explicitly positioned itself to adapt to an environment ripe for innovation in insect proteins and plant-based proteins.

A simple, well-articulated purpose such as Maple Leaf's can be transformative. In the classic 1960 HBR article "Marketing Myopia," Ted Levitt famously explained that the reason railway companies went out of business was that they thought of themselves as railway companies. Had they seen themselves as transportation companies or mobility companies, they might have found a second life. As the article states, the key to a good articulation of the desired end state lies in focusing on the fundamental role that the company plays in its system rather than the products it currently makes.

Once a company has identified its desired end state, it must then reach out to partners throughout the business system to win their buy-in to its new vision. In the case of Maple Leaf, executives and managers worked with their trade association, the Meat Institute, to create the Protein PACT to promote sustainable agriculture practices among farmers, ranchers, packers, and processors



across North America. The CEO of the Meat Institute has commended Maple Leaf's "partnership and generosity in sharing its expertise with peer processor companies of all sizes for the advancement of the whole industry."

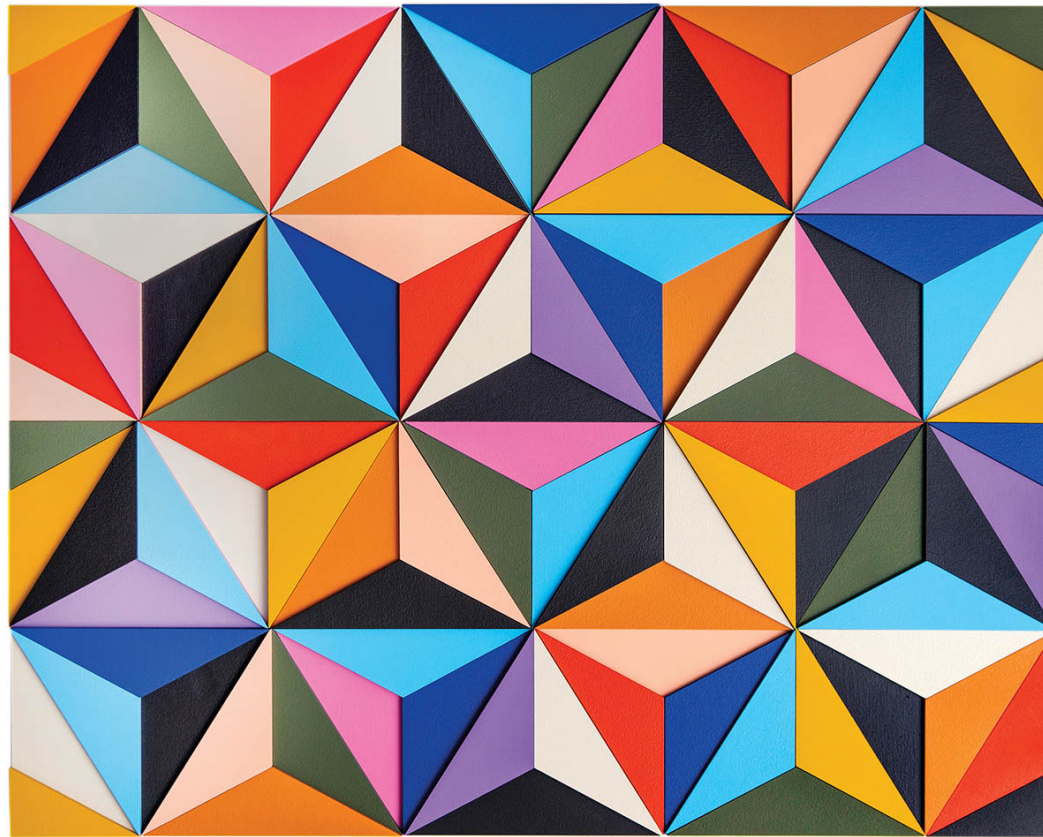
In the course of their outreach, companies should expect resistance or, perhaps more commonly, indifference, as a company's desired end state may seem irrelevant to some stakeholders. That brings us to the next principle.

2. Frame the problem, reframe it, and repeat. Breakthrough thinkers and design thinkers invest time in identifying the right problem and then fixate on it until it is solved. Systems thinkers recognize that there is often no single way to define a complex problem and that they'll need to reframe the definition iteratively to engage stakeholders who may experience a system's dysfunctions differently. The trick is to find out how problems that your ecosystem partners are experiencing relate to the problem you are trying to solve. In only a very few cases will you find that wicked problems like sustainability don't affect your system partners.

Consider our experience advising the University of Guelph. Located in Ontario, one of Canada's agricultural heartlands, the university was seeking to attract more grants to carry out research



in regenerative agriculture, in which producers strike a balance between efficient farming and sustainable practices. Leaders at the university believed they could best attract funding by showing the impact of their work. They asked us to help them incubate new ventures (involving Guelph researchers, farmers, and local entrepreneurs) that applied agricultural technologies to tackling climate change.



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As we embarked on the work, we realized that although farmers cared about climate change, the issue was not top of mind for them. We had difficulty engaging them because they had more urgent day-to-day challenges, such as planting, harvesting, and ensuring strong yields. We decided to reframe the problem from climate change to soil health. The reasoning was that soil health is more central to farmers' daily reality while still being deeply related to climate change. Healthy soil requires rich and diverse populations of microbes. Climate change causes soil to degrade, making it less effective at capturing carbon and at supporting biodiversity, among other harmful effects. Those effects in turn exacerbate climate change in a vicious feedback loop. In other words, industrial farming experiences the cost of climate change through its impact on soil quality. If farmers understood the relationship between soil quality and climate change, we figured, they would be more interested in the university's research.

With this reframing, more farmers answered our calls and participated in the workshops with other stakeholders. The work resulted in a number of innovations, from low-tech solutions, such as offering sheep for hire to serve as natural grazers and fertilizers, to high-tech ones, such as developing and applying iron oxide nanoparticles to the soil. The initiative also served to validate and accelerate the deployment of Susterre Technologies' high-pressure water system for planting seeds, allowing farmers



to maintain crop cover, which sequesters more carbon than tilling soil does.

3. Focus on flows and relationships, not products or services.

Most innovators focus their energy on finding a product or service that will solve the problem they've identified. And from a user's perspective, it is much easier to make sense of the innovation when it comes in the form of a physical object—for example, Meta's new smart glasses or Braun's integration of internet-enabled technologies into its electric toothbrushes.

However, an innovation need not be a new product, service, or feature to solve a problem. Changes to the flows or relationships among actors can be just as effective, either by reducing friction to speed things up or adding friction to slow down some parts of the system.

Consider insurance companies, which face increasing costs as property claims rise owing to the growing magnitude and frequency of climate-change-related weather events. The contractors that insurance companies hire often throw damaged materials into landfills and rebuild houses with new materials that draw down natural resources. All of this contributes to even more climate change.



Co-operators, a leading financial services and insurance company in Canada, knew that it had to disrupt the flow of damaged materials to landfills and improve the flow of restored materials back into homes. It developed a resiliency and sustainability strategy and identified specific initiatives that would help its clients and communities rebuild in ways that would reduce waste and prevent future losses. For example, Co-operators introduced “drying in place” for homes that experience damage from clean water. Instead of stripping out and replacing drywall and floorboards, vendors bring in high-powered fans to remove moisture from houses, preventing mold problems and allowing owners to return to their homes within 72 hours. What’s more, through its new “soft contents” cleaning process, Co-operators can also restore owners’ furniture, clothes, and other possessions, thereby saving items of sentimental value and reducing materials sent to the landfill.

Systems thinkers reframe the definition of a complex problem iteratively to motivate stakeholders who may experience a system’s dysfunctions differently.



Co-operators built sustainability requirements into its vendor-selection process to ensure that when homeowners experience a claims event, they can more easily choose a vendor that has embedded sustainability principles and practices into its operations. By removing frictions related to choosing sustainable solutions, Co-operators has been able to save money, protect the environment, and improve customer satisfaction.

It's important to note that Co-operators didn't create any new products, nor did it make huge investments. Rather, it worked creatively to improve the quality of flows and relationships within its existing business system. This is fundamental to systems thinking and differentiates it from other approaches to innovation.

4. Nudge your way forward. A lot of the rhetoric on innovation focuses on immediate solutions: the moon shot, the silver bullet, the killer app. Such solutions, however, usually create knock-on problems. Systems-thinking innovators seek to create an “ecology of actions” that steadily addresses problems within a system. They look not for leaps but for nudges and experiments that reveal insights into the system and move it forward, until it reaches a tipping point and evolves naturally.



Of course, design thinking shares this emphasis on behavior and experimentation, but it retains nonetheless a focus on a specific solution. Systems-thinking experiments are about exposing interdependencies among partners in the ecosystem as opposed to testing whether a product or service improves the user experience.

Consider the CSA Group, whose purpose is to advance safety, social good, and sustainability for Canadians through standards development. The benefits of better standards are sizable. According to the United Nations Environment Programme, about 37% of global carbon emissions are associated with the built environment, of which 38% could be reduced through “circular” building techniques. CSA embarked on a research project that applied circular design principles to the built environment, such as adaptive reuse of buildings and putting discarded material from buildings back into the economy. There are many players in the built environment, including architects, engineers, developers, owners, and material manufacturers. Standards are needed to get them to talk to one another so that circular design becomes embedded in every step of the process.

In working with CSA, we suggested that rather than try to brainstorm its way to a set of new, better standards that supported its North Star of creating a circular built environment, the



organization should focus on an ecology of small actions that covered three areas: educating architects, engineers, developers, and owners about the possibilities of a circular built environment; developing new standards that permitted reuse of materials; and finding ways to unlock small amounts of funding for new pilot projects. We also recommended that CSA engage a “coalition of the willing”—activists that had already bought into the idea of a circular built environment. It should launch a few pilot projects, and with each successful effort, it could bring in more people and more projects, with each step helping the community move closer to circularity.

Often the participants in a circular ecosystem have numerous ideas about what needs to be done, but the complexity is so daunting that they don’t know where to begin. The actions we recommended were a way to nudge things forward, allowing participants to understand the system better and then map out subsequent actions.

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In advocating for systems thinking we do not expect that companies or entrepreneurs will abandon the current dominant approaches to innovation. To begin with, breakthrough thinking is essential to technological progress. And design thinking is very



effective in bringing managers closer to their customers. But when issues are complex and affect an array of stakeholders in very different ways, those methods may create as many problems as they solve. Meeting the wicked challenges that society is experiencing is almost certainly beyond the capabilities of even the most inspired breakthrough or design thinkers. In these situations, systems thinking provides a more robust framework for change.

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